EXPERIMENT NO.1

Aim: To study the MATLAB environment and familiarize with basic MATLAB commands used for signal generation, manipulation, visualization, and system analysis.

CODE:

```
clc;
clear;
close;
%simple arithmatic operations
a=10; b=5;
disp("addition="+string(a+b));
disp("subtraction="+string(a-b));
disp("multiplication="+string(a*b));
disp("division="+string(a/b));
t=0:0.001:2*3.14;
w=-0:1:20;
x=sin(t);
y = cos(w);
subplot(2,1,1);
plot(t ,x);
xlabel("time");
ylabel("amplitude");
title("continous time sinewave");
subplot(2,1,2);
stem(w,y);
xlabel("time");
ylabel("amplitude");
title("discrete time cosine wave");
```

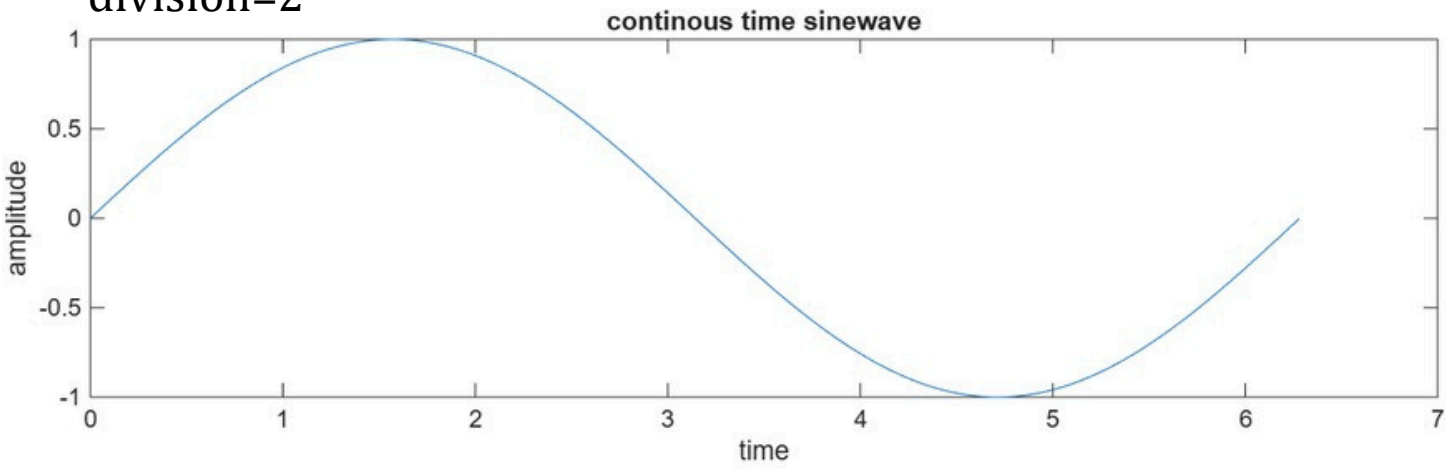
OUTPUT:

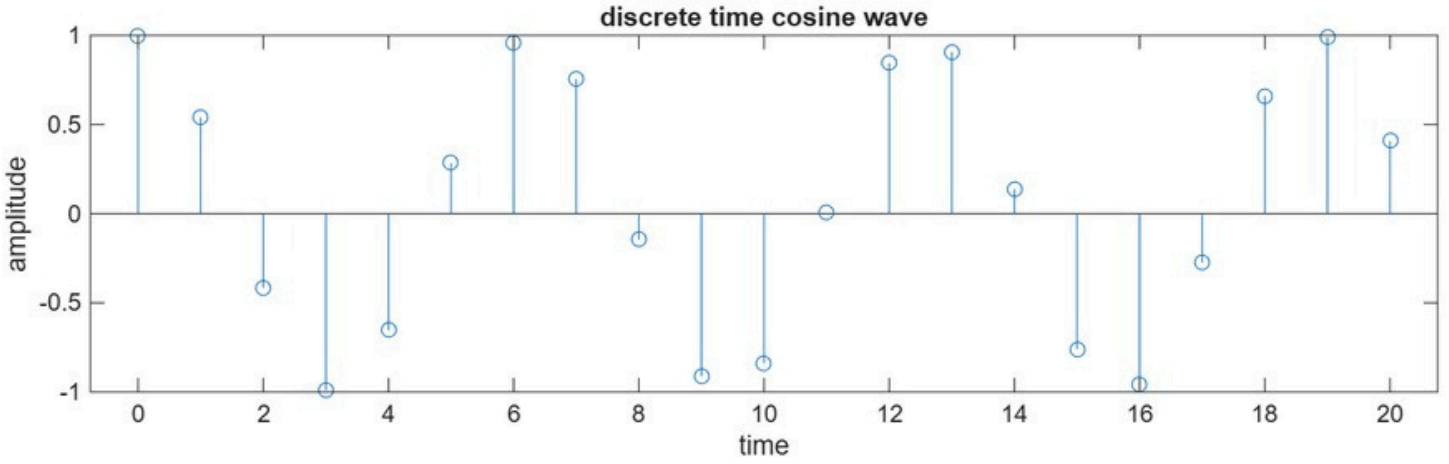
addition=15

subtraction=5

multiplication=50

division=2





EXPERIMENT NO.2 :

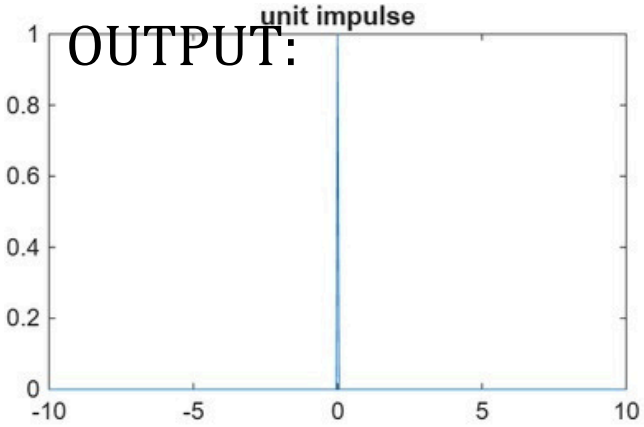
Generate and plot basic signals – Impulse, Unit Step, Ramp, Signum using MATLAB/Scilab. Further generate and plot various complex signals using time shifting, scaling and time reversal properties.

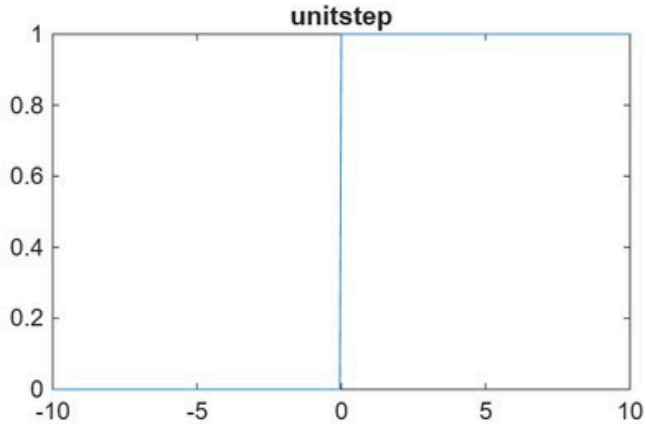
CODE:

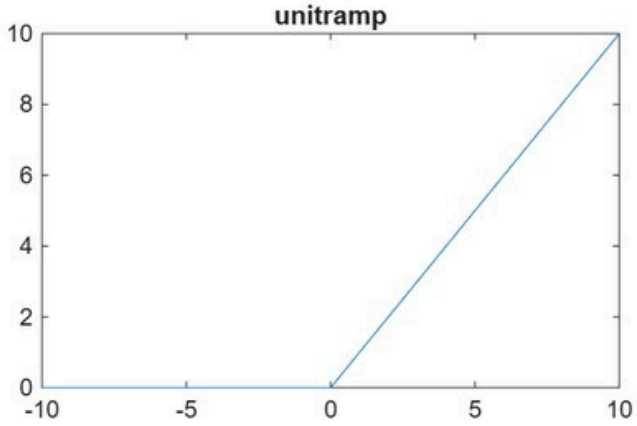
```
clc;
clear;
close;
%basic signal
t=-10:0.05:10;
impulse=zeros(1,length(t));
[m,idx]=min(abs(t));
impulse(idx)=1;
%unit step
ustep=zeros(1,length(t));
ustep((t>=0))=1;
%unit ramp
r=t .*ustep;
%signum
s=sign(t);
%plotting of graph
figure();
subplot(2,2,1);
plot(t,impulse);
title("unit impulse");
subplot(2,2,2);
plot(t ,ustep);
title("unitstep");
subplot(2,2,3);
plot(t ,r);
title("unitramp");
```

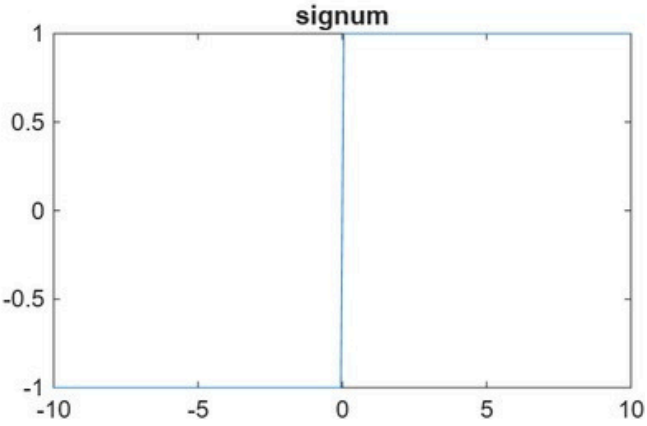
```
subplot(2,2,4);
plot(t ,s);
title("signum");
% part 2 shifting of signal
%time shifting shifted unit step(t-2))
ushift=zeros(1,length(t)); ushift((t-
2>=0))=1;
%time shifting shifted unit step(t+2))
ushift1=zeros(1,length(t));
ushift1((t+2>=0))=1;
%unit ramp
r=(t-2).*ushift;
p=(t+2).*ushift1;
figure();
subplot(2,3,1);
plot(t ,ustep);
title("unitstep");
subplot(2,3,4);
plot(t ,r);
title("unitramp");
subplot(2,3,2);
plot(t ,ushift);
title("shift unit step by (t-2)")
subplot(2,3,3);
plot(t ,ushift1);
title("shift unit step by (t+2)")
subplot(2,3,5);
plot(t ,r);
title("shifted ramp signal by (t-2) ")
subplot(2,3,6);
plot(t,p);
title("shifted ramp signal by (t+2)")
```

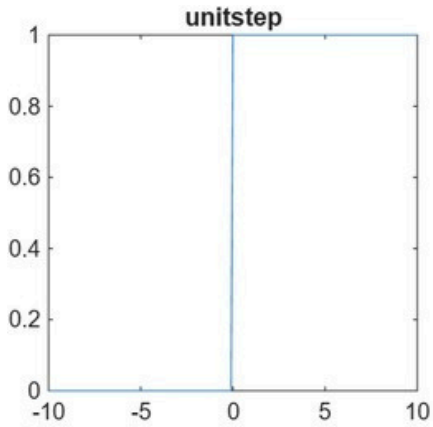
OUTPUT:

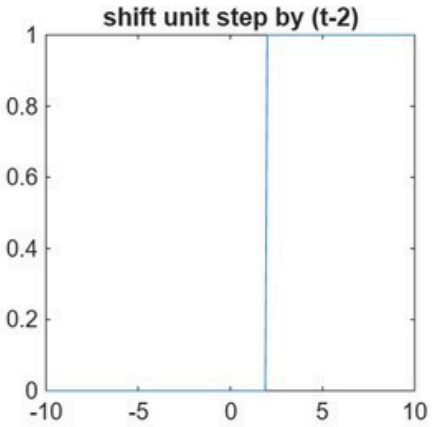

unit impulse
1
0.8
0.6
0.4
0.2
0
-10
-5
0
5
10

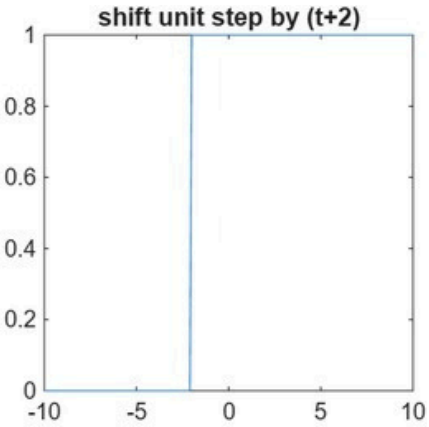

unitstep
1
0.8
0.6
0.4
0.2
0
-10
-5
0
5
10

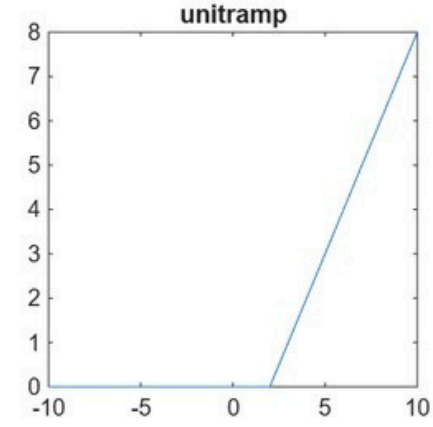

unitramp
10
8
6
4
2
0
-10
-5
0
5
10

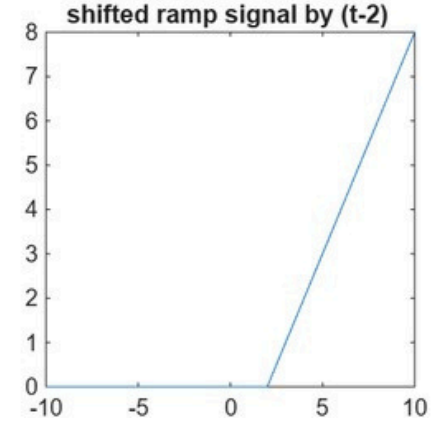

signum
1
0.5
0
-0.5
-1
-10
-5
0
5
10

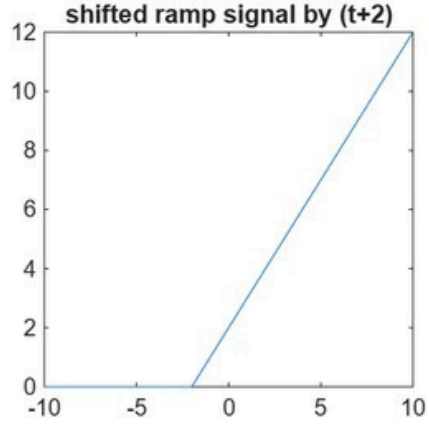

unitstep
1
0.8
0.6
0.4
0.2
0
-10
-5
0
5
10


shift unit step by (t-2)
1
0.8
0.6
0.4
0.2
0
-10
-5
0
5
10


shift unit step by (t+2)
1
0.8
0.6
0.4
0.2
0
-10
-5
0
5
10


unitramp
8
7
6
5
4
3
2
1
0
-10
-5
0
5
10


shifted ramp signal by (t-2)
8
7
6
5
4
3
2
1
0
-10
-5
0
5
10


shifted ramp signal by (t+2)
12
10
8
6
4
2
0
-10
-5
0
5
10

EXPERIMENT NO.3 :

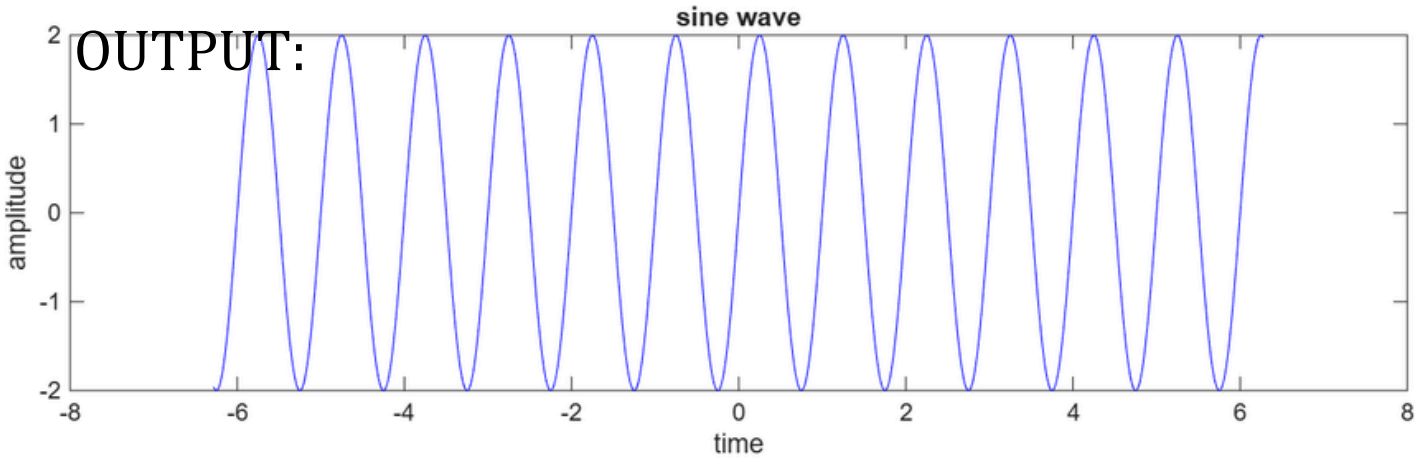
Generate the following continuous time domain signal and plot using MATLAB /Scilab (i) Sine/Cosine signal (ii) Sinc function (iii) Triangle waveform (iv) Rectangular Pulse (v) Sawtooth Waveform (vi) Exponential function

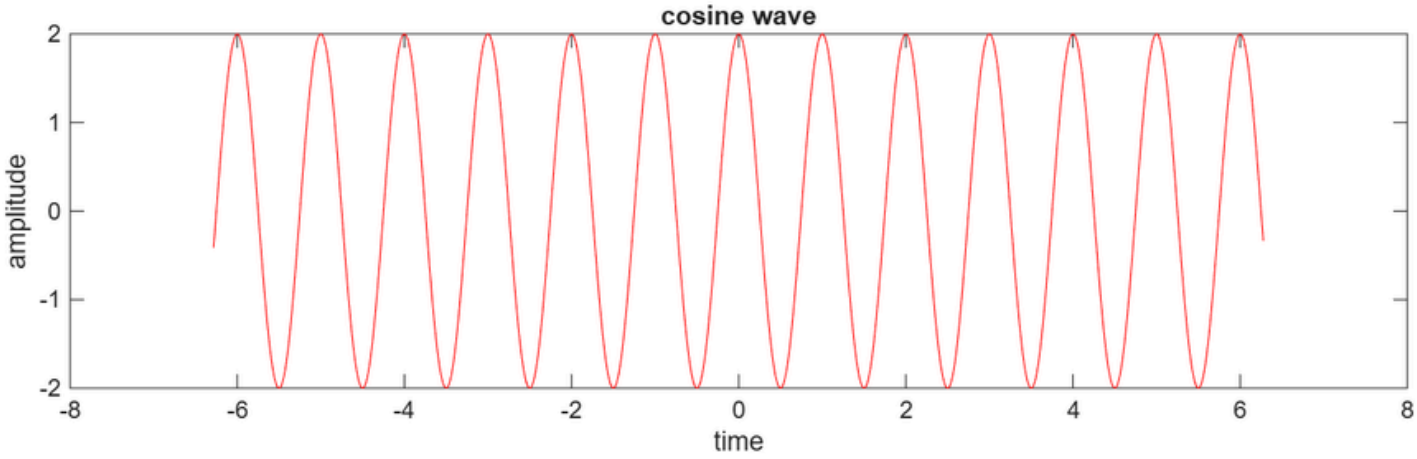
CODE:

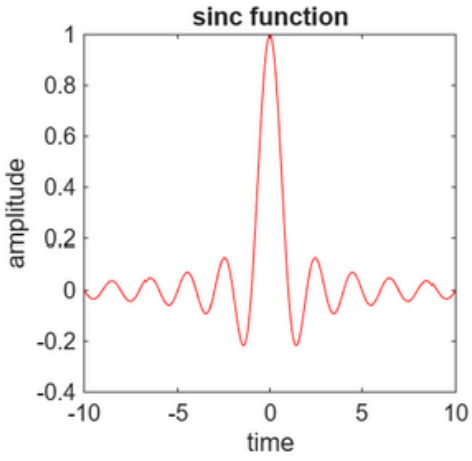
```
clc;
clear;
close;
t=-2*pi:0.01:2*pi;
A=2;
f=1;
x1=A*sin(2*pi*f*t);
x2=A*cos(2*pi*f*t);
subplot(2,1,1);
plot(t,x1,'b');
xlabel("time");
ylabel("amplitude");
title("sine wave");
subplot(2,1,2);
plot(t,x2,'r');
xlabel("time");
ylabel("amplitude");
title("cosine wave");
figure();
t1=-10:0.01:10;
x3=sin(pi*t1)./(pi*t1);
x3((t1==0))=1;
subplot(2,3,1);
plot(t1,x3,'r');
xlabel("time");
ylabel("amplitude");
```

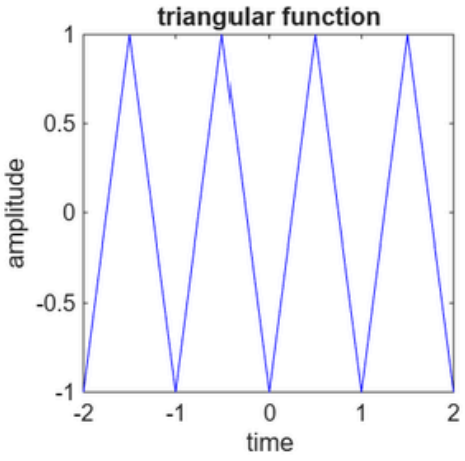
```
title("sinc function");
t2=-2:0.001:2;
x4=2*abs(2*(t2-floor(t2+0.5)))-1;
subplot(2,3,2);
plot(t2,x4,'b');
xlabel("time");
ylabel("amplitude");
title("triangular function");
t3=-2:0.001:2;
x5=(2*(t3-floor(t3+0.5)));
subplot(2,3,3);
plot(t3,x5,'r');
xlabel("time");
ylabel("amplitude");
title("sawtooth function");
t4=0:0.001:5;
a=-1;
x6=exp(a*t4);
subplot(2,3,4);
plot(t4,x6,'b');
xlabel("time");
ylabel("amplitude");
title("decaying exponential function");
t5=0:0.001:5;
a=1.5;
x7=exp(a*t5);
subplot(2,3,5);
plot(t5,x7,'r');
xlabel("time");
ylabel("amplitude");
title("increasing exponential function");
```

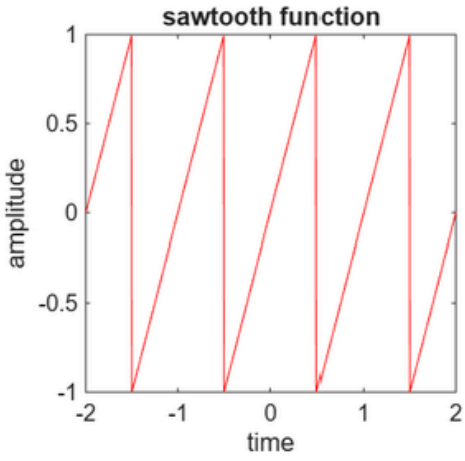
OUTPUT:

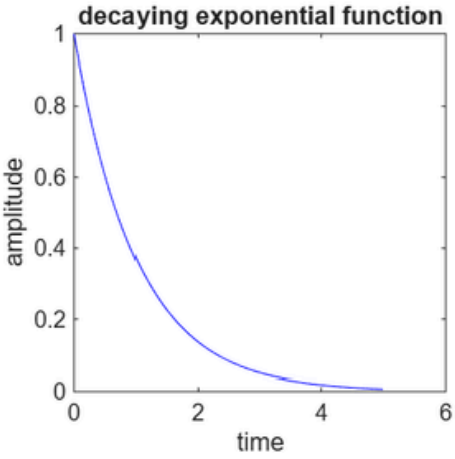

sine wave
amplitude
time

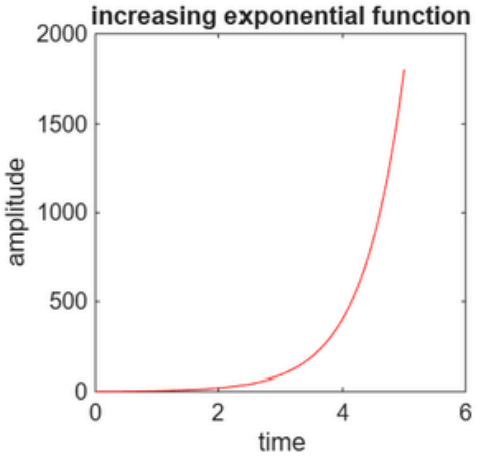

cosine wave
amplitude
time


sinc function
amplitude
time


triangular function
amplitude
time


sawtooth function
amplitude
time


decaying exponential function
amplitude
time


increasing exponential function
amplitude
time

EXPERIMENT NO.4 :

To study and verify the following properties of signal/system using MATLAB /Scilab (i) Linearity (ii) Stability (iii) Time invariant. Find the Energy and Power of the given signal. Separate Even and odd components of given signal and plot respectively.

CODE:

```
% 1.Write a matlab code to verify the discrete time signal is linear or
% non-linear.
clc;
close all;
clear all;
n=1:50;
x1=[1,0,-1,2];
x2=[1,-1,1,-1];
a=input('enter the value of multiplier of a:- ');
b=input('enter the value of multiplier of b:- ');
x=a*x1+b*x2;
y=2.^x ; %y=2^x
y1=2.^x1; %y1=2^x1
y2=2.^x2; %y2=2^x2
y_n=a*y1+b*y2 ;
figure();
subplot(2,1,1);
stem( y);
xlabel("sample");
ylabel("amplitude");
title("case 1");
subplot(2,1,2);
stem(y_n);
xlabel("sample");
ylabel("amplitude");
title("case 2");
tolerance=0.0001;
if(abs( y-y_n)<=tolerance)
 then
 disp('the system is linear');
else
 disp('the system is non linear');
end;
```

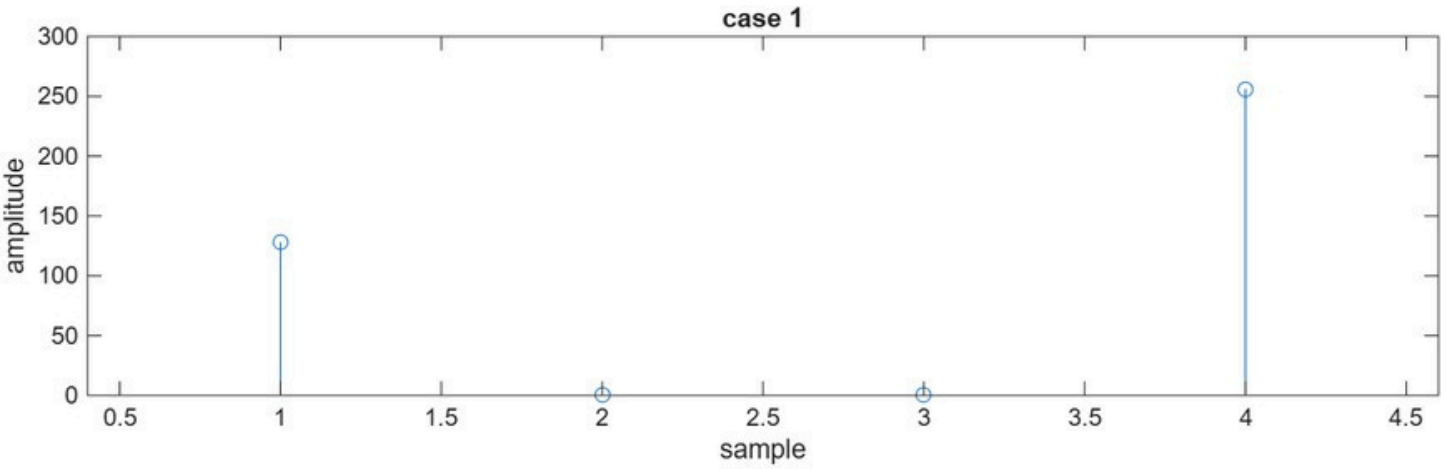
```
% time delay
x=[1,0,-1,2];
k=input('enter the value of delay:- ');
x3=[zeros(1,k) x];
y3=2.^x3; %y(n,k)(output with delayed input)
yt=2.^x;
figure();
subplot(2,1,1);
stem(y3);
xlabel("sample");
ylabel("amplitude");
title("output with delayed input");
y4=[zeros(1,k) yt]; %y(n-k)(delayed output);
subplot(2,1,2);
stem(y4);
xlabel("sample");
ylabel("amplitude");
title("delayed output");
tolerance=0.0001;
if(abs( y3-y4)<=tolerance)
   then
 disp('the system is time invariant');
else
 disp('the system is time variant');
end;
% 2.Write a matlab code to verify the discrete time signal is energy or not.
x4=[1,-1,2,-2,3];
E=sum(abs(x4).^2);
disp(E);
% stability
if(E<inf )
 disp('the system is stable');
else
 disp('the system is unstable');
end;
```

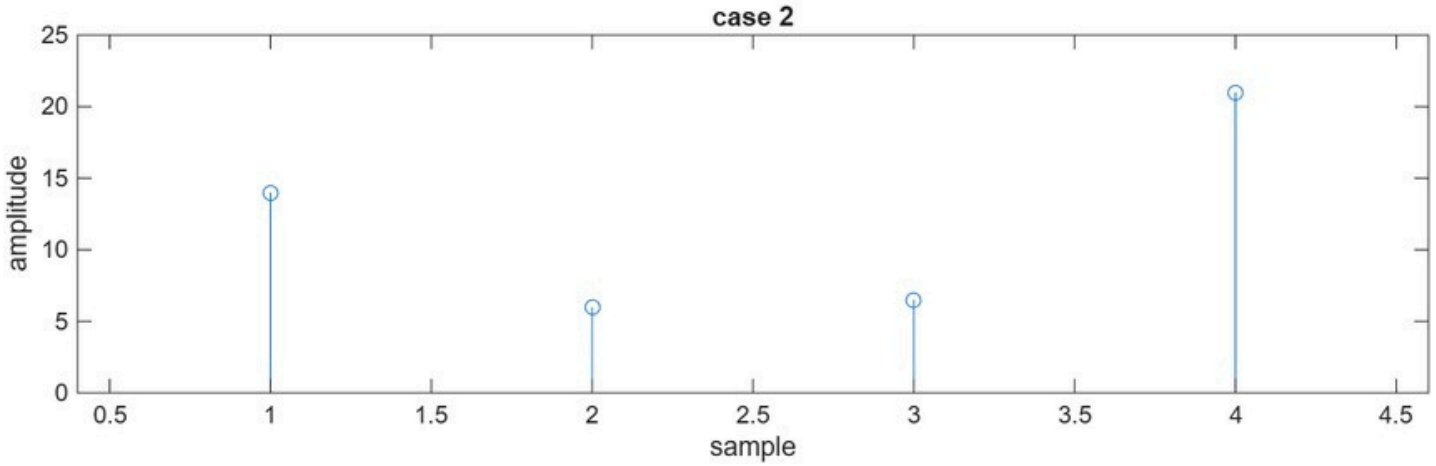
```
% 3.Write a matlab code to detect the even and odd component of the
% discrete time signal
figure()
x5=[2,-2,3,4];
xe=(x5+x5(end:-1:1))/2;
xo=(x5-x5(end:-1:1))/2;
subplot(1,3,1);
stem(x5);
xlabel("sample");
ylabel("amplitude");
title("original signal");
subplot(1,3,2);
stem(xe);
xlabel("sample");
ylabel("amplitude");
title("even component");
subplot(1,3,3);
stem(xo);
xlabel("sample");
ylabel("amplitude");
title("odd component");
```

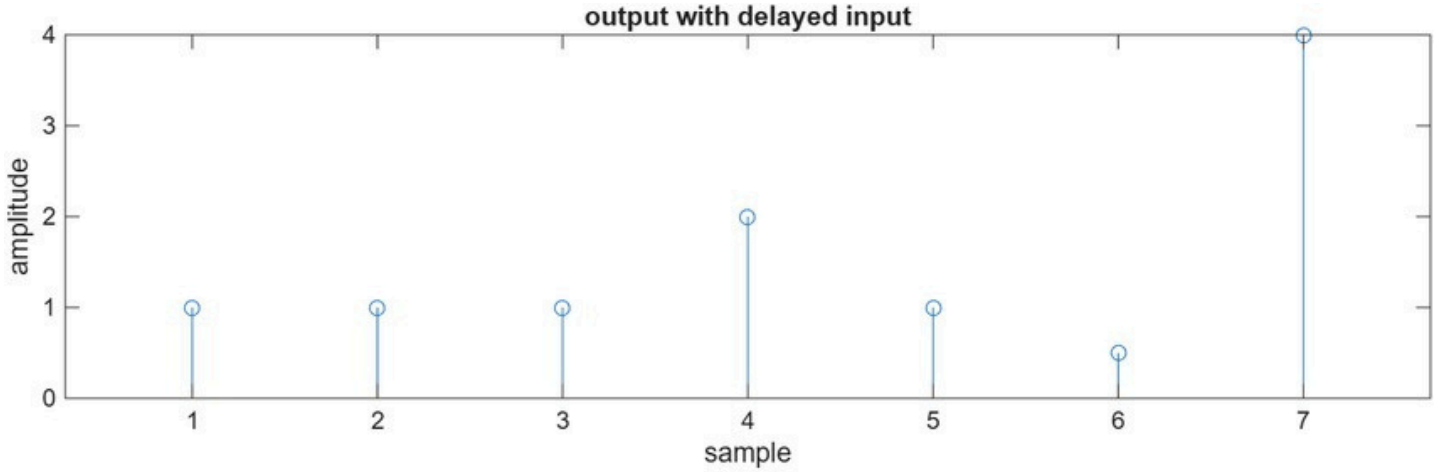
OUTPUT:

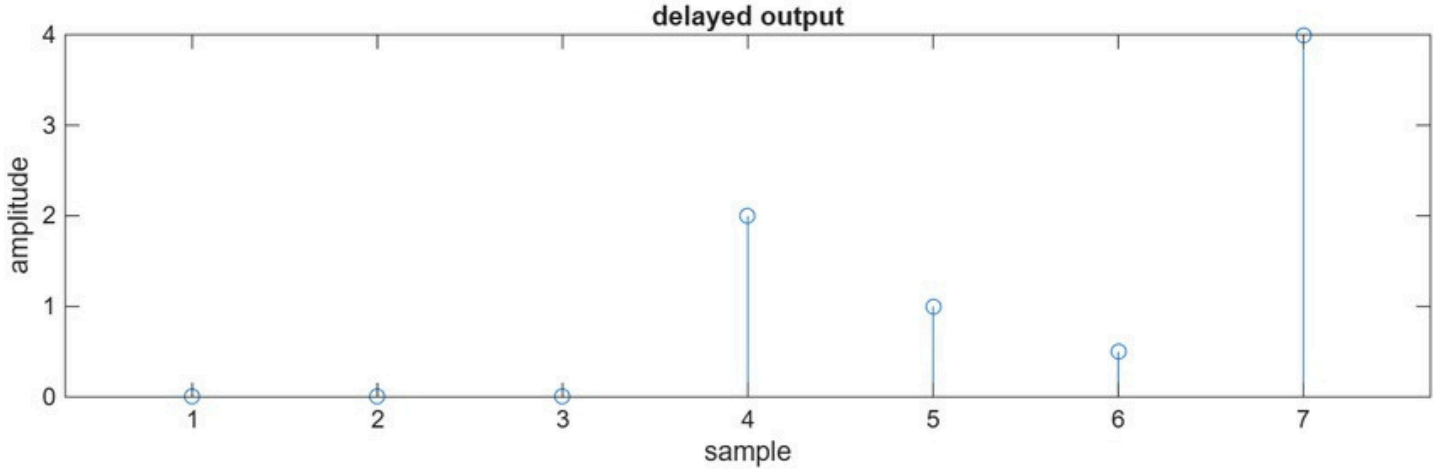
enter the value of multiplier of a:- 5
enter the value of multiplier of b:- 2
the system is non linear
enter the value of delay:- 3
the system is time variant

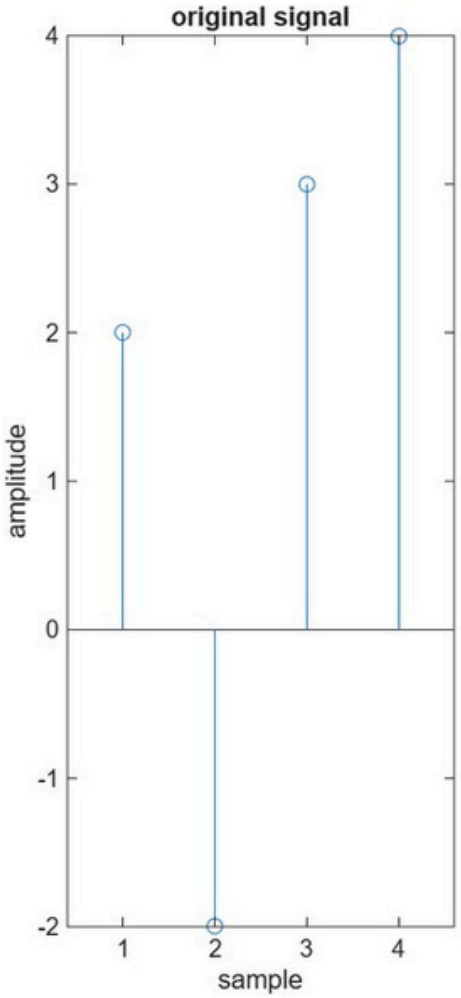
the system is stable

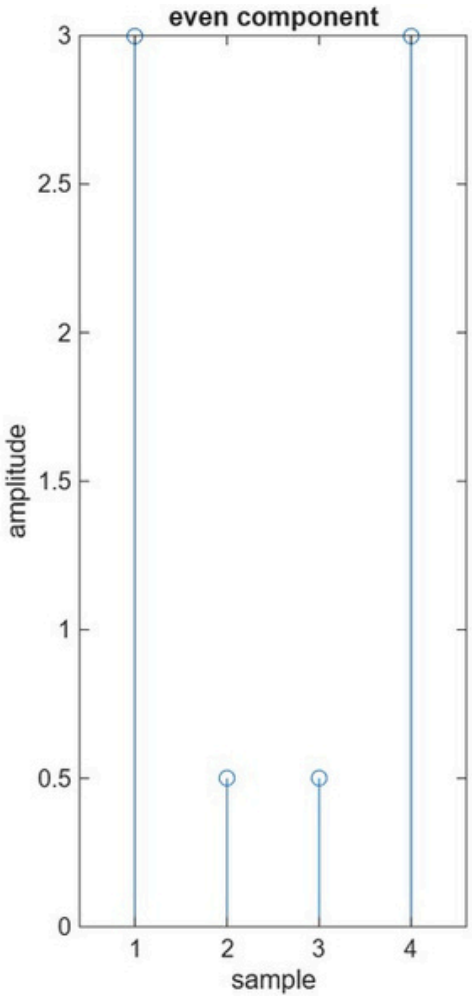

case 1
amplitude
sample
300
250
200
150
100
50
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5

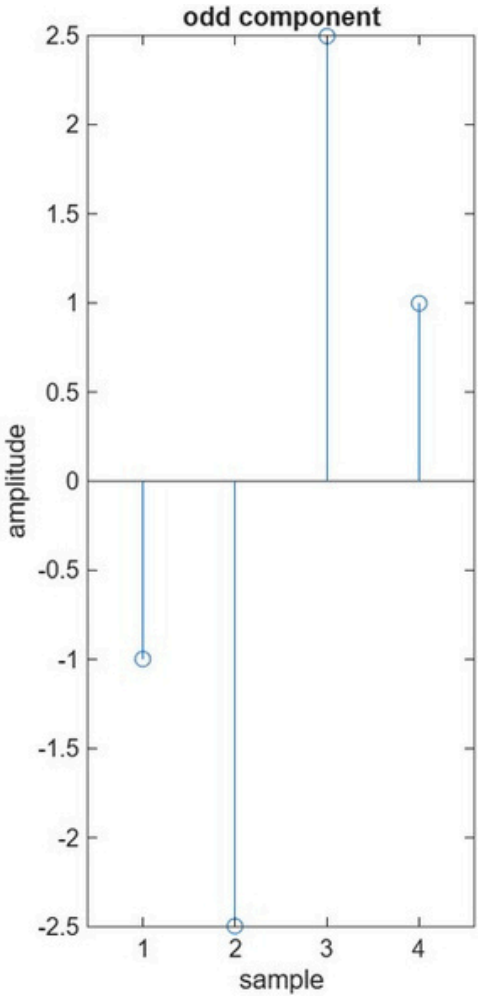

case 2
amplitude
sample
25
20
15
10
5
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5


output with delayed input
amplitude
sample
4
3
2
1
0
1
2
3
4
5
6
7


delayed output
amplitude
sample
4
3
2
1
0
1
2
3
4
5
6
7


original signal
amplitude
sample
4
3
2
1
0
-1
-2
1
2
3
4


even component
amplitude
sample
3
2.5
2
1.5
1
0.5
0
1
2
3
4


odd component
amplitude
sample
2.5
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-2.5
1
2
3
4

EXPERIMENT NO.5 :

To Study convolution of following different sequences using MATLAB/Scilab (without using conv command) and plot the resultant signal. i Convolution between two rectangular signals. ii Convolution between rectangular and Triangular signals. iii Convolution between two Triangular signals.

CODE:

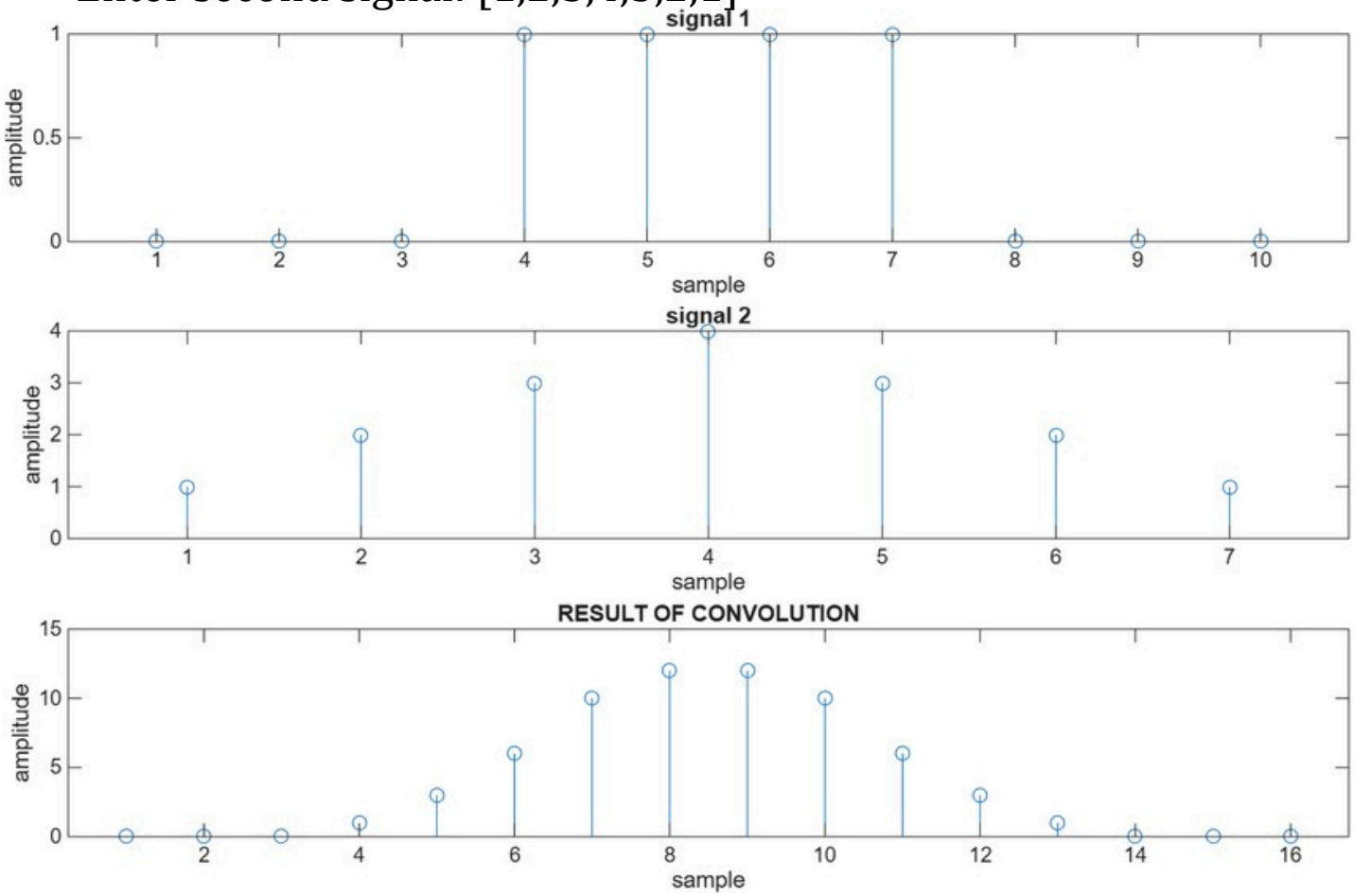
```
clc;
close all;
clear all;
function y = Conv_user(x,h)
l1 = length(x);
l2 = length(h);
l3 = l1+l2-1;
x1 = [x, zeros(1,l2)];
h1 = [h, zeros(1,l1)];
y = [zeros(1,l3)];
for n = 1:l3
   sum = 0;
   for k = 1:n
      sum = sum + x1(k)*h1(n - k + 1);
   end;
 y(n) = sum;
end;
end;
s1 = input("Enter first signal: ");
s2 = input("Enter second signal: ");
y = Conv_user(s1,s2);
figure;
subplot(3,1,1);
stem(s1);
xlabel("sample");
ylabel("amplitude");
title("signal 1");
```

```
subplot(3,1,2);
stem(s2);
xlabel("sample");
ylabel("amplitude");
title("signal 2");
subplot(3,1,3);
stem( y);
xlabel("sample");
ylabel("amplitude");
title('RESULT OF CONVOLUTION');
```

OUTPUT:

Enter first signal: [0,0,0,1,1,1,1,0,0,0]

Enter second signal: [1,2,3,4,3,2,1]



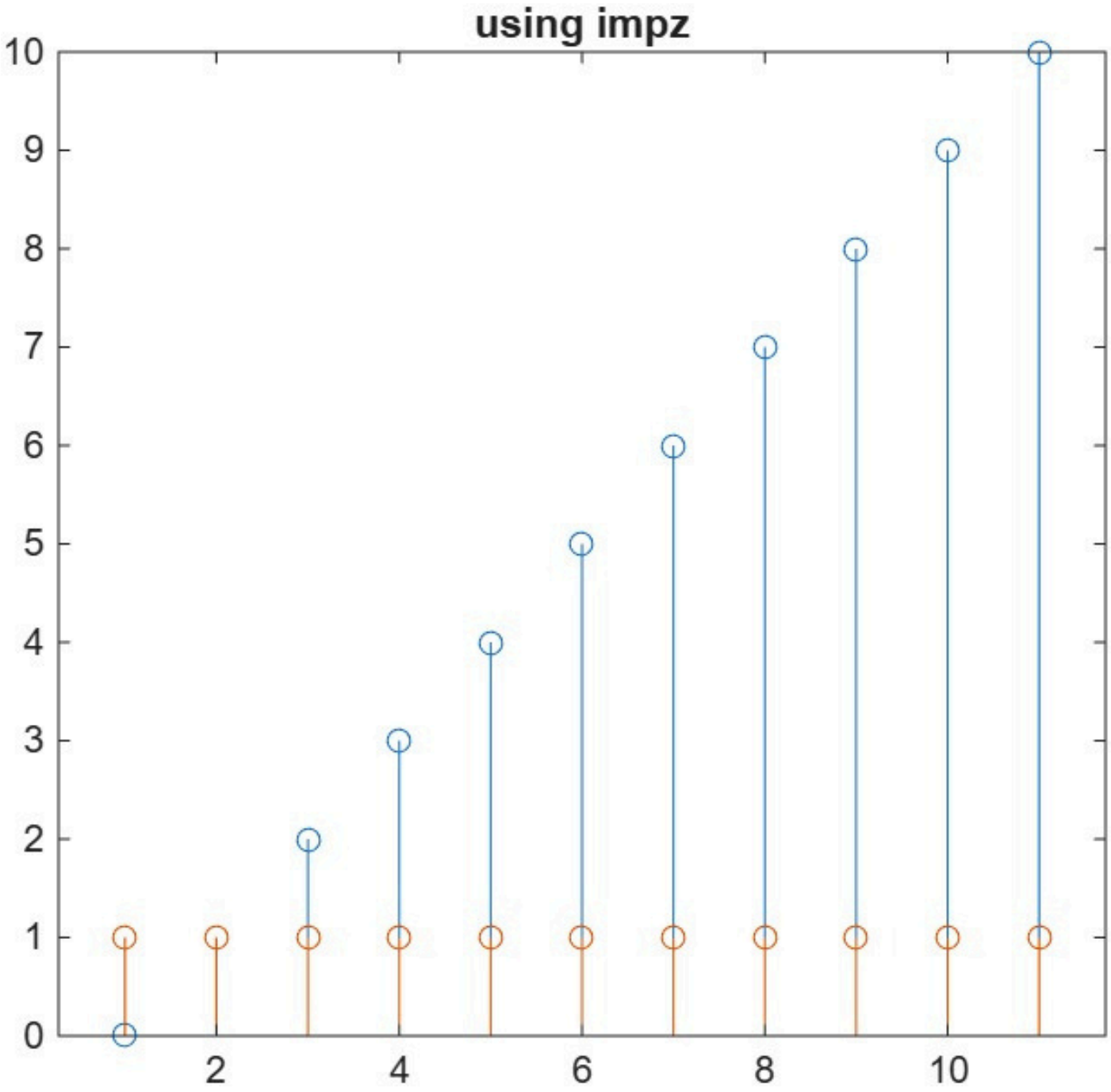
EXPERIMENT NO.6 :

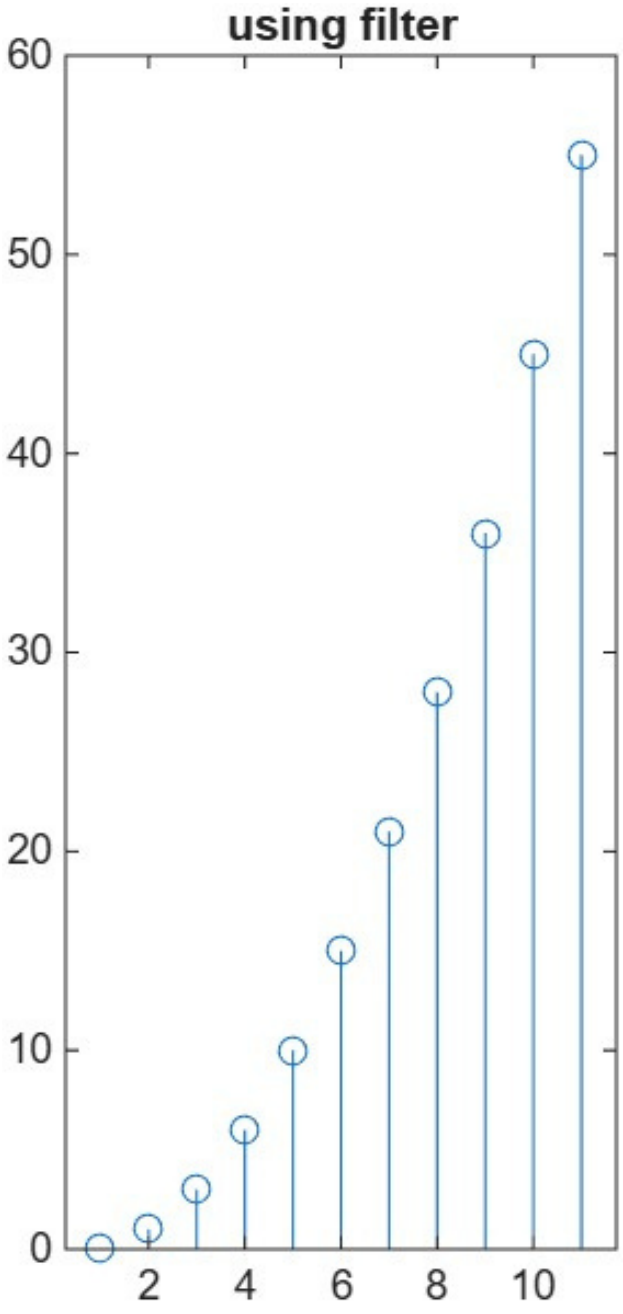
To study impz, filter and freqz properties of signal using MATLAB/Scilab.

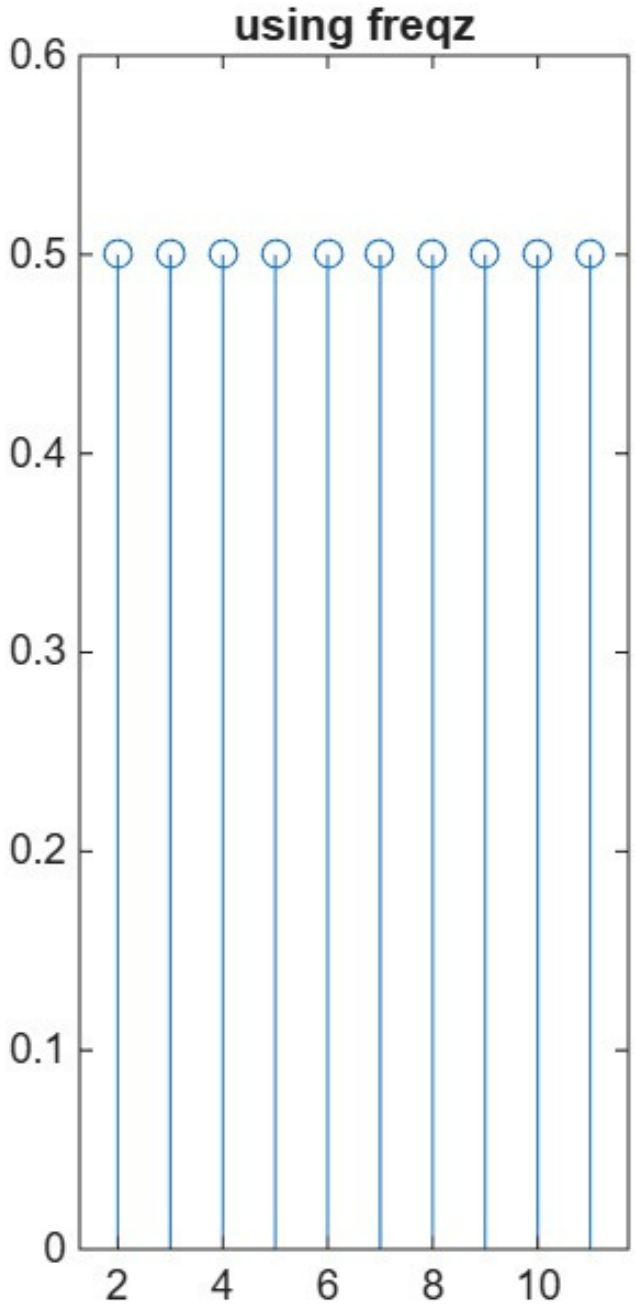
CODE:

```
clc;
clear all;
close all;
b=[1];
a=[1 -1];
n=0:1:10;
figure();
stem(n);
hold on;
stem(impz(b,a,n));
title("using impz");
figure();
subpot(1,2,1);
stem(n);
stem(filter(b,a,n));
title("using filter");
subplot(1,2,2);
stem(n);
stem(freqz(b,a,n));
title("using freqz");
```

OUTPUT:




using filter
60
50
40
30
20
10
0
2
4
6
8
10


using freqz
0.6
0.5
0.4
0.3
0.2
0.1
0
2
4
6
8
10

EXPERIMENT NO.7 :

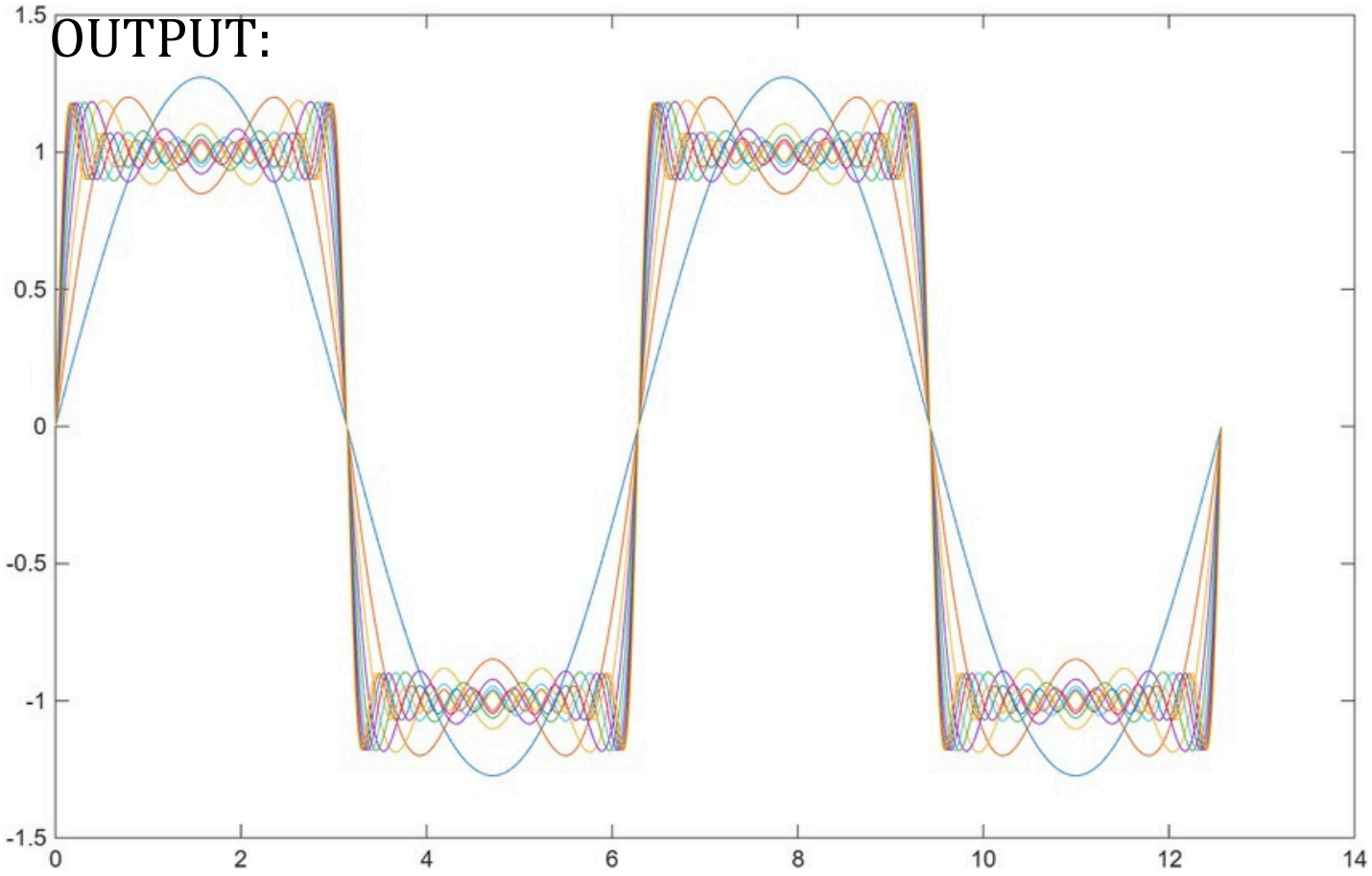
To Study the Fourier series of Square, Sawtooth and Triangular waves using MATLAB/Scilab.

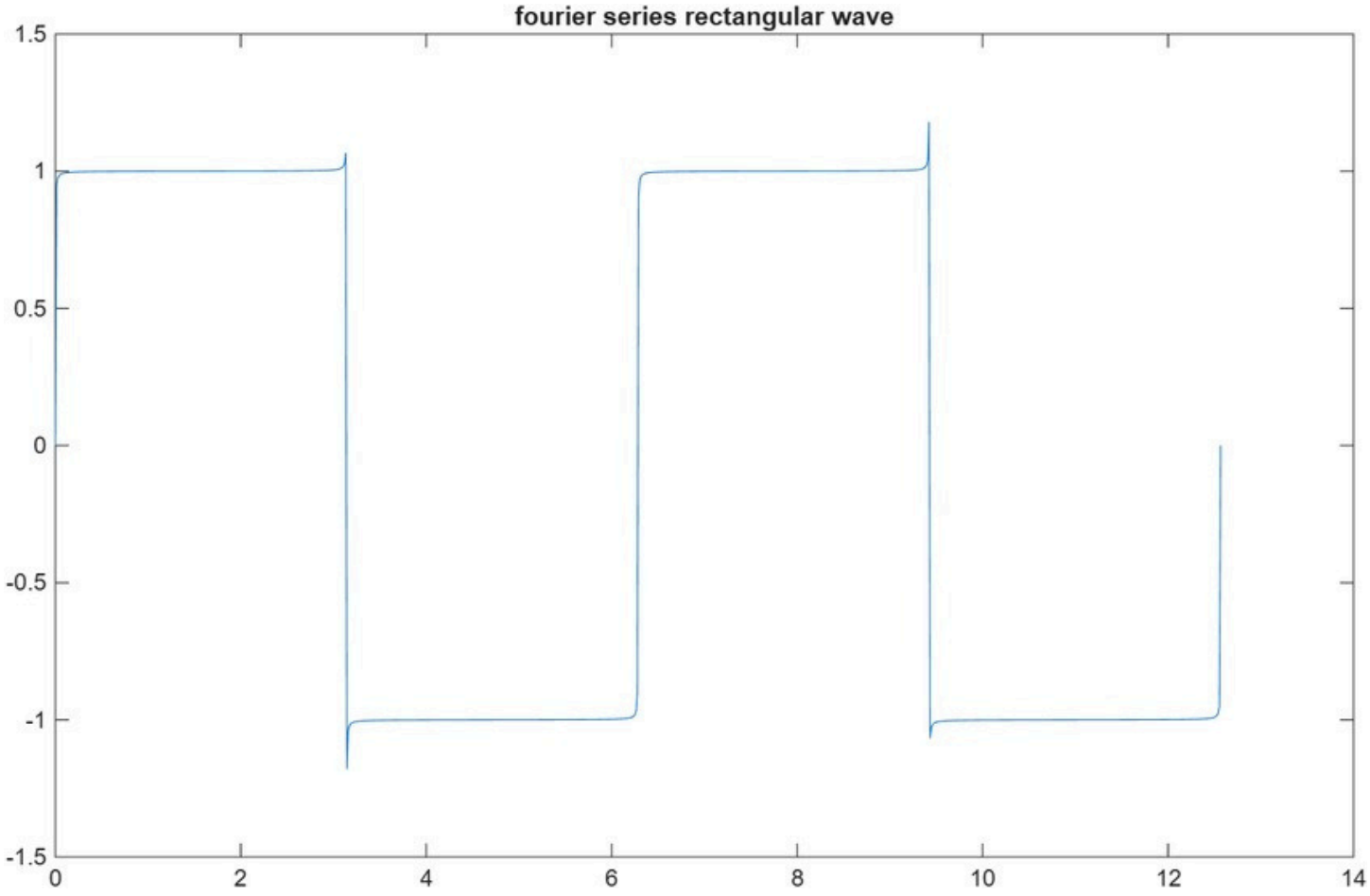
CODE:

```
clc;
close all;
clear all;
T = 2*pi;
omega = 2*pi/T;
t = linspace(0,4*pi,1000);
N = 1000; % Harmonics
sq_wave = zeros(size(t));
for n = 1:2:N
 sq_wave = sq_wave + 4/(n*pi)*sin(n*omega*t);
plot(t,sq_wave);

   hold on;
   pause(2);
end
figure();
plot(t ,sq_wave);
title("fourier series rectangular wave");
```

OUTPUT:


1.5
1
0.5
0
-0.5
-1
-1.5
0
2
4
6
8
10
12
14


fourier series rectangular wave
1.5
1
0.5
0
-0.5
-1
-1.5
0
2
4
6
8
10
12
14

EXPERIMENT NO.8 :

To Study Fourier Transform of a signal using FFT and IFFT command and plot magnitude and phase spectra of a signal using MATLA/LabVIEW.

```
clc;
clear;
close all;

% Define time and frequency vectors
t = -10:0.01:10;
w = -2*pi:0.01:2*pi;

% Define signals and their titles in a cell array
signals = {3.^t, 'Exponential Signal'; sin(t),'Sinewave'; t.^2,'Parabolic Wave' };

num_signals = size(signals, 1);

for i = 1:num_signals
 x_t = signals{i, 1};
 sig_title = signals{i, 2};

 % Vectorized Numerical Fourier Transform computation using trapz
 % Integral: X(w) = integral( x(t) * exp(-j*w*t) dt )
 X_k = trapz(t, x_t(:) .* exp(-1j * (t' * w)), 1);

 X_mag = abs(X_k);
 X_angle = angle(X_k);

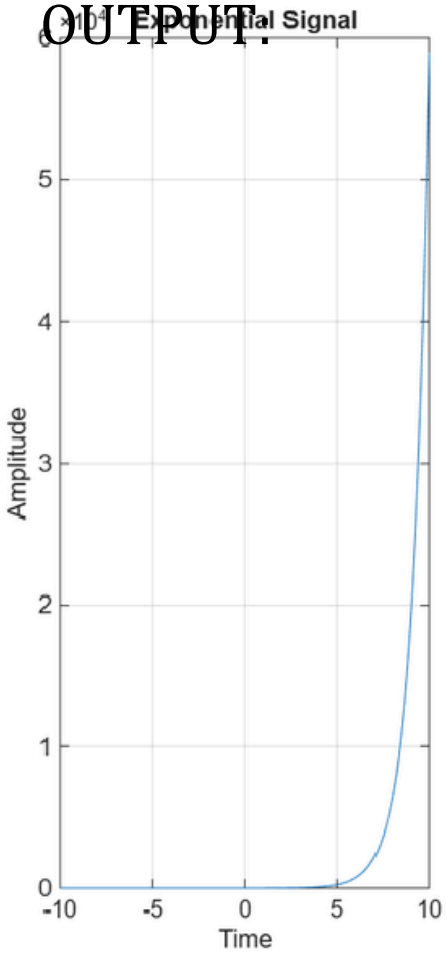
 % Create Figure
 figure('Name', sig_title, 'NumberTitle', 'off');

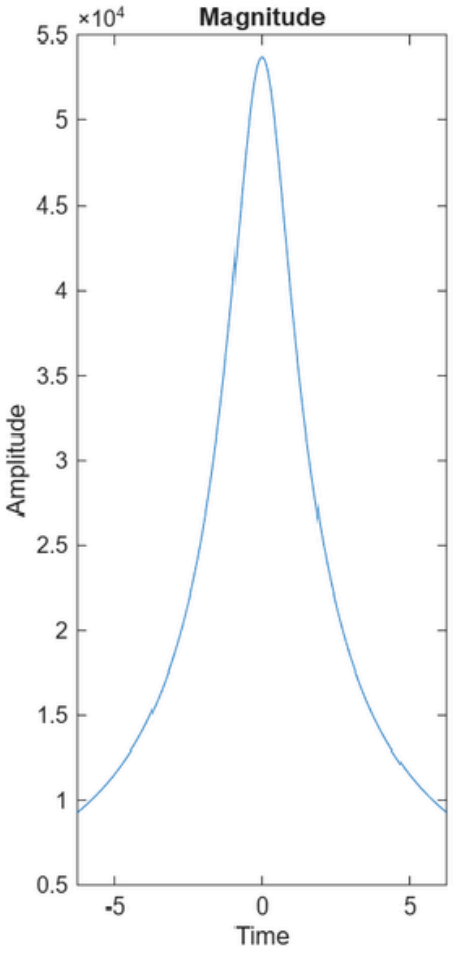
 % 1. Time Domain Plot
 subplot(1, 3, 1);
 plot(t, x_t, 'LineWidth', 1.2);
 title(sig_title);
 xlabel('Time (s)');
 ylabel('Amplitude');
 grid on;

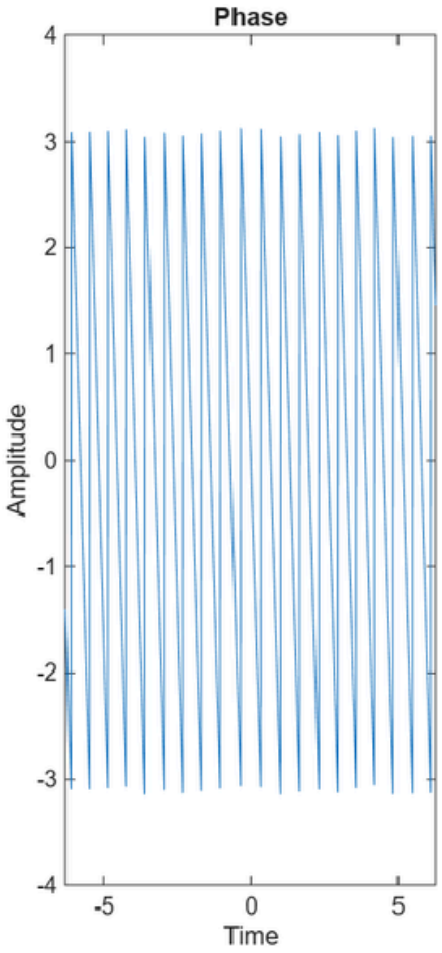
 % 2. Magnitude Spectrum
 subplot(1, 3, 2);
 plot(w, X_mag, 'LineWidth', 1.2);
 title(['Magnitude of ', sig_title]);
 xlabel('Frequency \omega (rad/s)');
 ylabel('Magnitude');
 grid on;

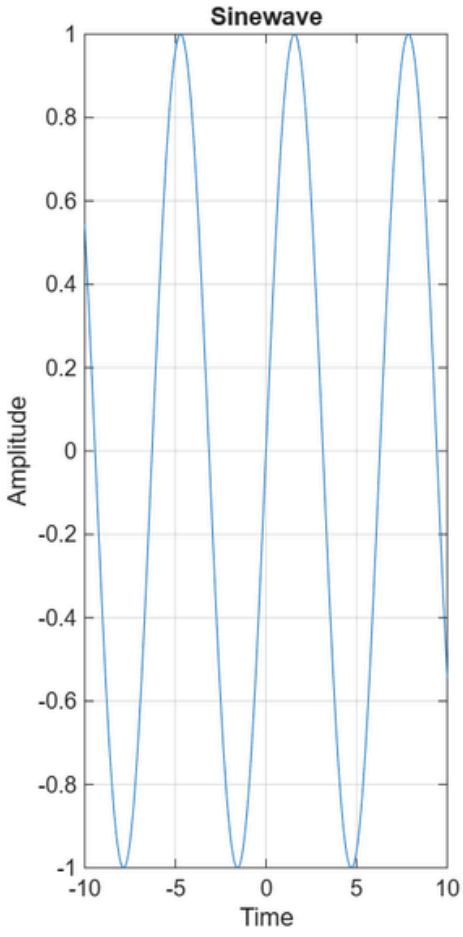
 % 3. Phase Spectrum
 subplot(1, 3, 3);
 plot(w, X_angle, 'LineWidth', 1.2);
 title(['Phase of ', sig_title]);
 xlabel('Frequency \omega (rad/s)');
 ylabel('Phase (rad)');
 grid on;
end
```

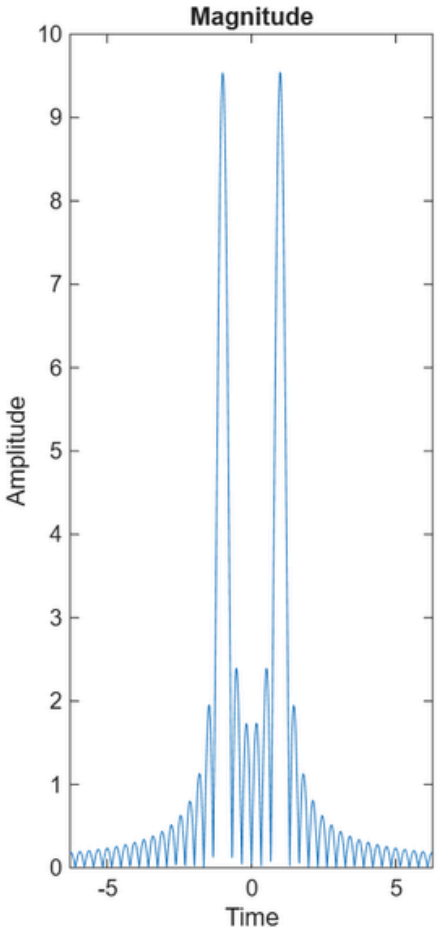
OUTPUT:

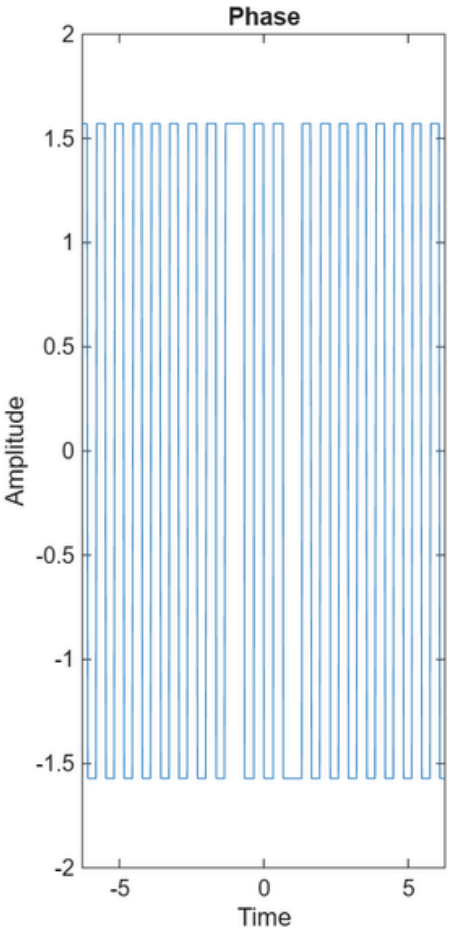

×10^4
Exponential Signal
Amplitude
Time

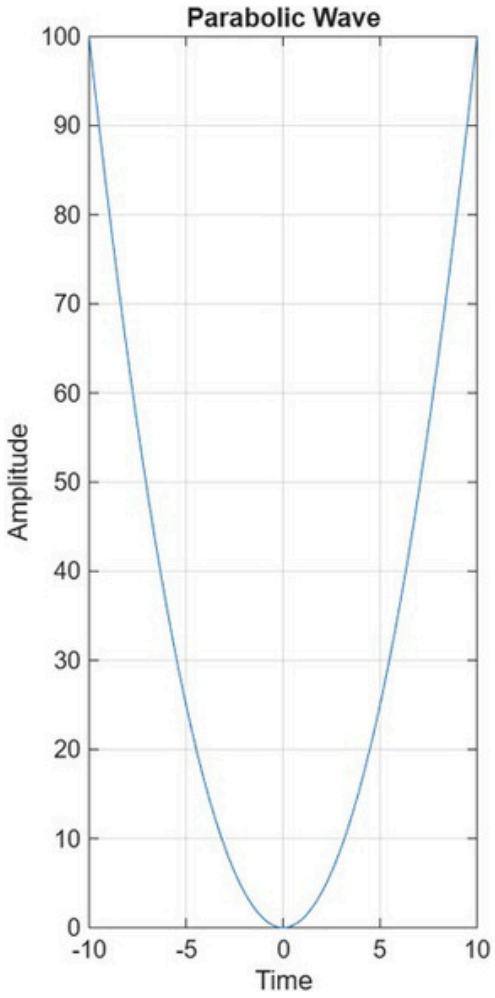

×10^4
Magnitude
Amplitude
Time

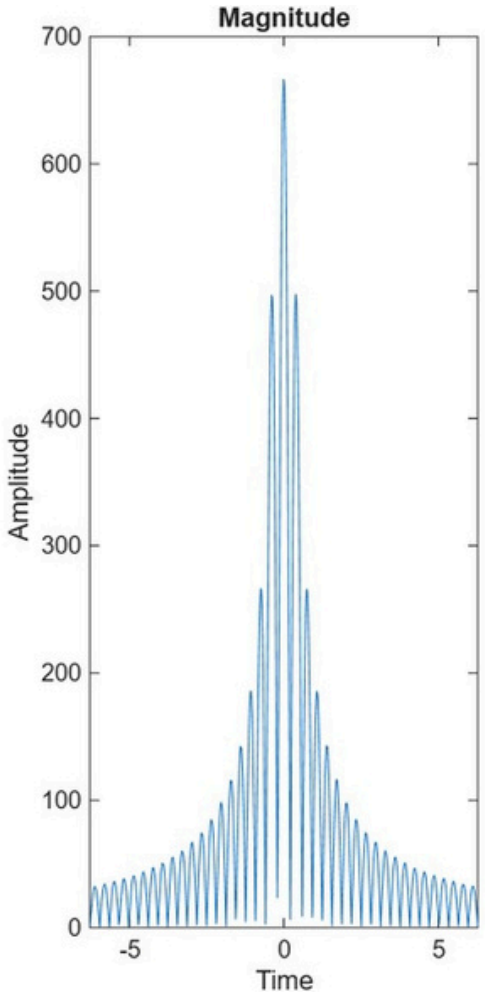

Phase
Amplitude
Time

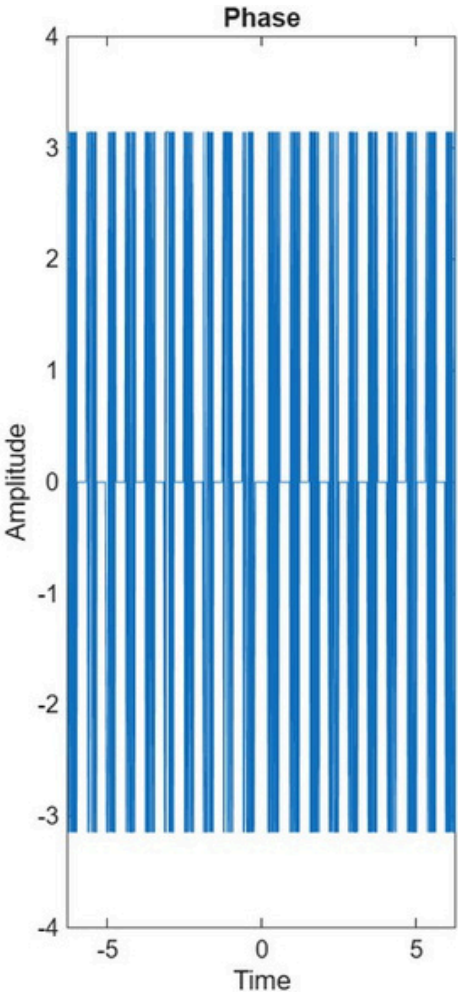

Sinewave
Amplitude
Time


Magnitude
Amplitude
Time


Phase
Amplitude
Time


Parabolic Wave
Amplitude
Time
100
90
80
70
60
50
40
30
20
10
0
-10
-5
0
5
10


Magnitude
Amplitude
Time
700
600
500
400
300
200
100
0
-5
0
5


Phase
Amplitude
Time
4
3
2
1
0
-1
-2
-3
-4
-5
0
5

EXPERIMENT NO.9 :

To Study Z Transform and plot region of Convergence using MATLAB Toolbox.

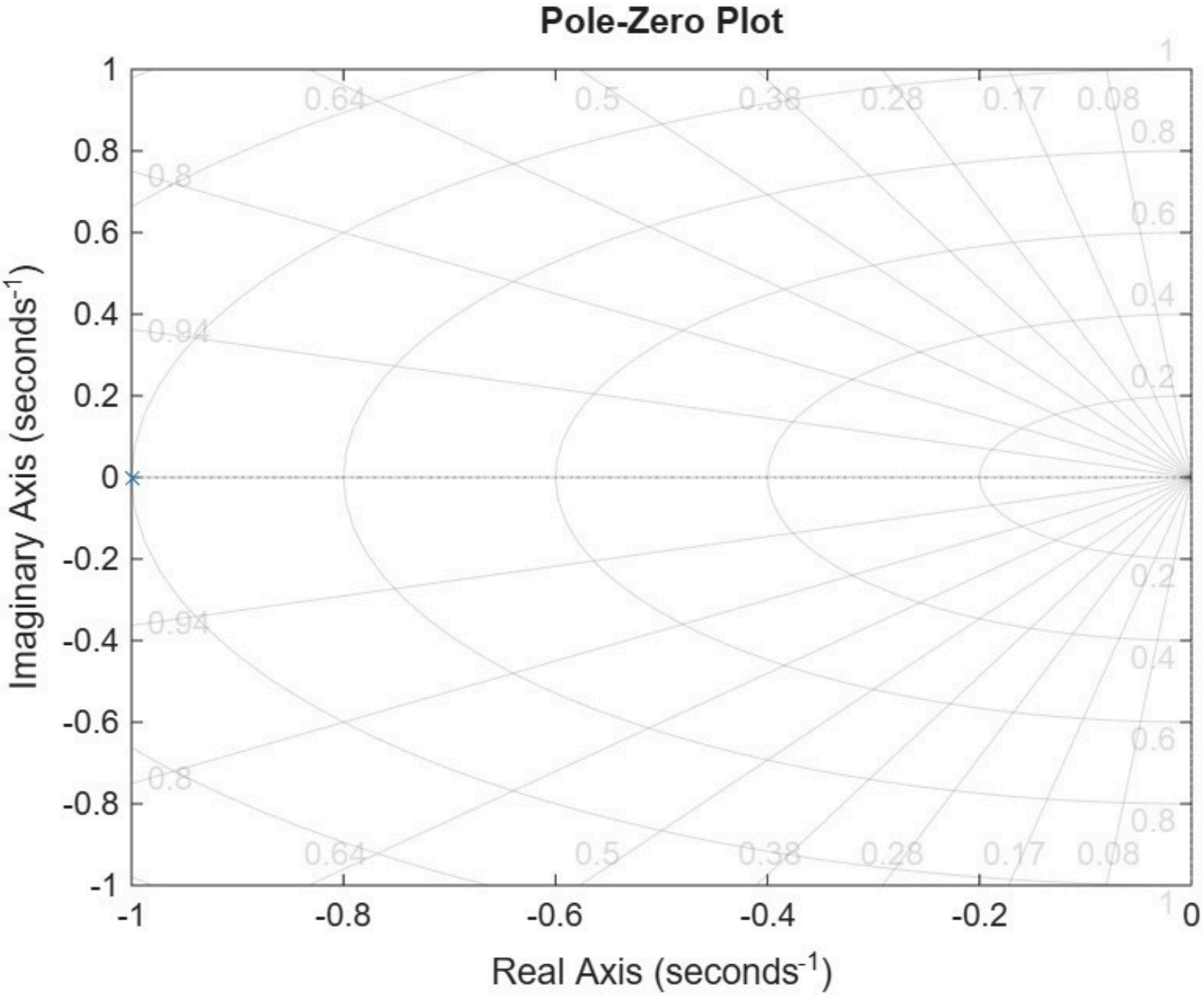
CODE:

```
clc;
clear all;
close all;
syms n;
x= sin(n);
f=ztrans(x);                disp('Z-
transform      of      sin(n):');
pretty(f );
inversex       =       iztrans(f);
disp('Inverse  Z-transform:');
pretty(inversex);
figure;
pz          =          tf([1],[1,2,1]);
pzmap(pz);
grid on;
title('Pole-Zero Plot');
```

OUTPUT:

```
Z-transform of sin(n):
z sin(1)
------------------
 2
z - cos(1) z 2 + 1

Inverse Z-transform:
sin(n)
```


Pole-Zero Plot
Imaginary Axis (seconds-1)
Real Axis (seconds-1)
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0.64
0.5
0.38
0.28
0.17
0.08
0.94